

and we are going to have lots of great prizes!

If you are generally excited about Ubuntu, check out <http://community.ubuntu.com> and you will quickly find something interesting to which you can contribute as well.

Q Ubuntu marked its retail presence in the country with Dell's laptops. The drive seems to have lost its sheen as Ubuntu did not partner with any other OEM for its retail presence in India. So what are the benefits you have derived from your retail presence in India?

Maria Bonnefon also told me that we have rolled out with Dell in 850 stores, and we are preparing for roll out with another major OEM in the upcoming months.

Another story we are very excited about is happening in Uttar Pradesh, where the government has sponsored 1.5 million Ubuntu/HP laptops to promote IT literacy. Big deployments were also realised in Delhi and Assam. In the state of Uttar Pradesh, one of the world's largest government-sponsored laptop programmes is taking place with 1.5 million Ubuntu/HP laptops delivered.

Uttar Pradesh has witnessed economic growth in the knowledge-based and IT intensive service sectors in India. They want to promote IT literacy, as a means to encourage

their graduating students to think about careers like software development. With two million software developers in India, Uttar Pradesh is keen for its young people to join their number.

The Government of Assam has given 28,000 Ubuntu laptops to learners this year alone. It runs an incentive programme for Grade 12 students to receive a free computer if they achieve more than 60 per cent in their final exams. Their value is explained by MK Yadava, former MD, Assam Electronics Development Corporation (AMTRON): "The Ubuntu-based PCs are of huge benefit to the students of Assam. They have all the educational programs they need – from word processing to spreadsheet creation – and it's all ready to go from Day One."

At the University of Delhi, which is placed third among India's university rankings, 65,000 Ubuntu HP laptops have been deployed on campus and are being used by students.

The feedback in all these successful programmes has been that Ubuntu is a preferred solution in Indian education because:

- It's secure and virus-free.
- An abundance of education applications are available on the Ubuntu Software Centre.
- It has all the productivity software that students need in LibreOffice.
- It's low-cost and has been proven. 

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```
fs0:\EFI> cd boot
fs0:\EFI\boot> dir
Directory of fs0:\EFI\boot

09/12/13 06:47p <DIR>          1,024 .
09/12/13 06:47p <DIR>          1,024 ..
09/12/13 06:47p             2,040 HELLO.EFI
09/12/13 06:47p       1,049,496 bootldr.efi
09/12/13 06:22p         84,520 gnu\bootldr.efi
09/12/13 06:32p         38,880 osx2_x64.efi
09/12/13 06:32p        29,394 osx4_x64.efi
09/12/13 06:32p        30,504 iso9660_x64.efi
09/12/13 06:32p         43,456 rebasefs_x64.efi
                7 File(s)      1,257,440 bytes
                2 Dir(s)

fs0:\EFI\boot> HELLO.EFI

UEFI did programming demo. IC3 IBC back later.
http://jeleco.com.ua/
Press any key...
```

Figure 2 : Sample UEFI-application that outputs a text string

You can see the EFI Shell prompt here, so proceed further as follows:

"fs0:", "cd \Directory\Where\Your\UEFI\Apps\Stored" and execute a UEFI-application.

Despite the obvious scepticism, today we proved that UEFI and assembly language could fit each other perfectly. Several facts should be remembered though. EFI IA32 implementation uses a stack frame to pass all parameters through it, whilst x64 UEFI implements a slightly different model—the first four parameters are passed through

processor registers RCX, RDX, R8 and R9, and only subsequent arguments come through the stack frame.

When you address different pointer tables, it should be noted that for IA32 EFI architecture, the pointer size is 4 bytes (32 bits), and for x64 UEFI it comes as 8 bytes (64 bits). This fact must be taken into account during the calculation of a pointer address inside EFI system tables.

However, all these differences with various addressing problems for multiple hardware platforms can be omitted by designing a single, unified and cross-platform UEFI application based on EFI byte code concept. But more about this in our next article. 

References

- [1] <http://itassembly.net/download.php>
- [2] <http://wiki.gemu.org/Download>
- [3] <http://sourceforge.net/projects/edk2/files/OVMF/>

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